

DOCUMENTACIÓN TÉCNICA

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Matricula	2025-0773
Asignatura	Seguridad en Redes
Script	AshleyFabian_2025-0773_STP_RootClaim_P1.py
Practica	P1
Fecha	05 de June de 2026
Herramienta	Python 3 + Scapy
Plataforma	GNS3 — Kali Linux

1. Objetivo del laboratorio

Demostrar que Kali Linux (MAC 00:0c:29:66:a3:89) puede reclamar el rol de Root Bridge en la topología STP enviando BPDUs con prioridad 0. SW1 cambio su Root ID a la MAC de Kali con prioridad 0, y su puerto Gi0/1 (hacia Kali) paso a ser el Root Port. SW2 reconvergió apuntando a SW1 como mejor camino hacia el root.

- Enviar BPDUs con prioridad 0 para reclamar Root Bridge.
- Verificar reconvergencia en SW1: Root ID cambia a MAC de Kali.
- Verificar en SW2 el nuevo camino hacia el root.
- Aplicar BPDU Guard como contra-medida: puerto entra en err-disabled.

2. Objetivo del script

El script AshleyFabian_2025-0773_STP_RootClaim_P1.py construye Configuration BPDUs con prioridad 0 y Path Cost 0, usando la MAC de eth0 (00:0c:29:66:a3:89) como Bridge ID. Los envia al multicast STP 01:80:C2:00:00:00 cada segundo para mantener el rol de root.

2.1 Parametros

Parametro	Descripcion	Valor usado
-i / --iface	Interfaz de red	eth0
-c / --count	Cantidad de BPDUs (0=continuo)	continuo
-d / --delay	Hello time en segundos	1
-p / --priority	Prioridad del bridge	0

2.2 Requisitos

- Kali Linux con Python 3 y Scapy.
- Interfaz eth0 conectada a SW1 Gi0/1 (VLAN 10).
- STP habilitado en los switches (configuracion por defecto).
- Ejecutar como root.

2.3 Uso

```
sudo python3 AshleyFabian_2025-0773_STP_RootClaim_P1.py -i eth0

# Verificar en SW1
SW1# show spanning-tree vlan 10
```

3. Funcionamiento del script

- Obtiene MAC real de eth0 con get_if_hwaddr(): 00:0c:29:66:a3:89

- Construye BPDU: Ethernet -> LLC (dsap=0x42) -> STP con prioridad 0 y pathcost 0.
- Envía al multicast STP 01:80:C2:00:00:00 cada segundo.
- SW1 recibe BPDU con Bridge ID menor al suyo y reconverge.
- El script estaba activo antes de la demostración, por eso SW1 ya mostraba a Kali como root.

3.1 Resultados obtenidos

Metrica	Estado normal	Durante el ataque
Root Bridge SW1	SW1 (prio 32778, 0c0d.8a96.0000)	Kali (prio 0, 000c.2966.a389)
Root Port SW1	Ninguno (era root)	Gi0/1 (hacia Kali)
Root Bridge SW2	SW1 (prio 32778)	SW1 (camino hacia Kali)
Tras BPDU Guard	—	Gi0/1 err-disabled; SW1 vuelve a ser root

4. Documentacion de la red

Dispositivo	Rol	Interfaz / Puerto	IP / VLAN
CSR1000v	Router / Gateway	Gi2.10 — Gi2.20	Gi2.10: 192.168.73.1/24 Gi2.20: 192.168.74.1/24
SW1 — vIOS L2	Switch capa 2	Gi0/0 trunk R1 Gi0/1 VLAN10 Kali Gi0/2 trunk SW2	VLANs 10 (ATACANTE) y 20 (VICTIMA)
SW2 — vIOS L2	Switch capa 2	Gi0/0 VLAN20 VPCS Gi0/1 trunk SW1	VLANs 10 (ATACANTE) y 20 (VICTIMA)
Kali Linux	Atacante	eth0	192.168.73.10/24 — VLAN 10
VPCS	Victima	eth0	192.168.74.20/24 — VLAN 20

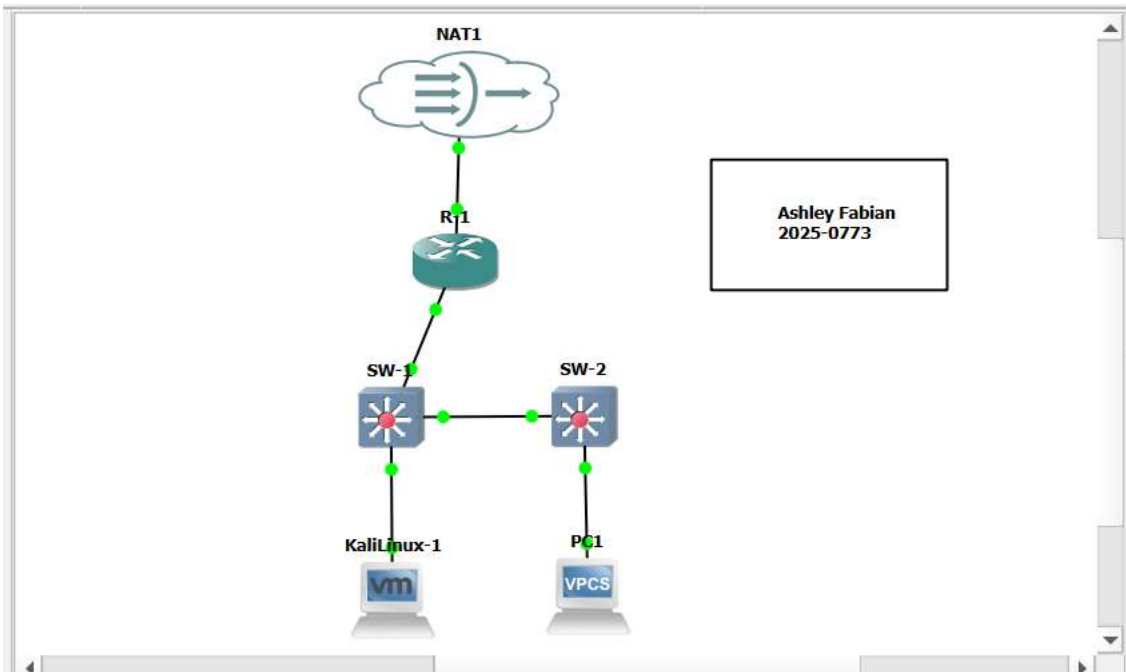
4.1 Estado STP verificado en laboratorio

```
# SW1 DURANTE el ataque:
SW1# show spanning-tree vlan 10
# Root ID Priority: 0 Address: 000c.2966.a389 (Kali)
# Bridge ID Priority: 32778 Address: 0c0d.8a96.0000 (SW1)
# Gi0/1 Role: Root FWD

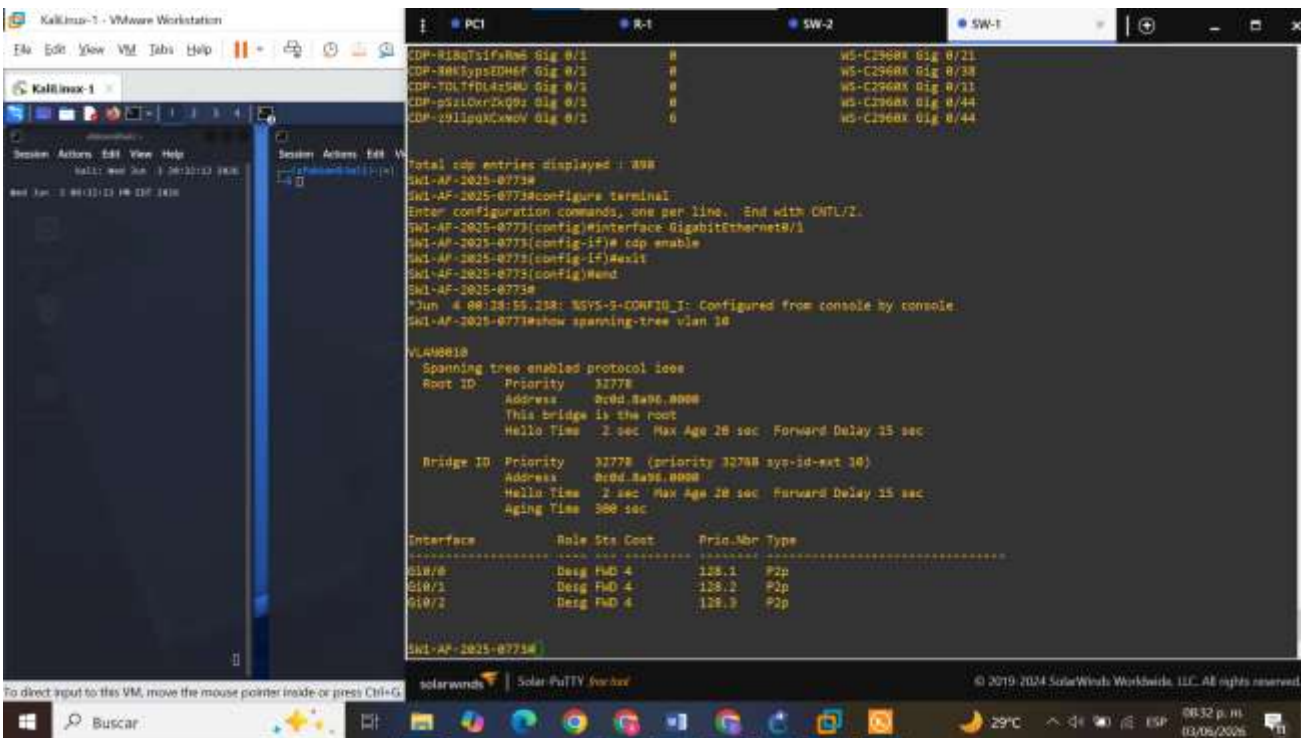
# SW1 TRAS BPDU Guard:
# Root ID Priority: 32778 Address: 0c0d.8a96.0000
# This bridge is the root
# Gi0/1: err-disabled
```

5. Capturas de pantalla

Topología con nombre y matrícula visibles:



Muestra del switch antes de realizar el ataque:



[illegible]


```

# configure interface
Jun  4 00:14:50.908: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
Jun  4 00:14:57.381: %VLAN-5-CONFIG_I: Configured from console by console
Jun  4 00:14:57.885: %LINE-5-UPDOWN: Interface GigabitEthernet0/1, changed state to down
SW1-AF-2025-0773#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     8c0d.8a36.0000
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID   Priority    32778 (priority 32768 sys-id-ext 10)
             Address     8c0d.8a36.0000
             Hello Time 1 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                     Desg Fwd 4       128.1   P2p
Gi0/2                     Desg Fwd 4       128.3   P2p

SW1-AF-2025-0773#show interfaces GigabitEthernet0/1 status

Port      Name             Status      Vlan    Duplex  Speed Type
Gi0/1     err-disabled10   err-disabled 10       auto    auto  10Gbps

SW1-AF-2025-0773#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1-AF-2025-0773(config)#interface GigabitEthernet0/1
SW1-AF-2025-0773(config-if)#spanning-tree portfast
SW1-AF-2025-0773(config-if)#spanning-tree bpdguard enable
SW1-AF-2025-0773(config-if)#exit
SW1-AF-2025-0773(config)#end
SW1-AF-2025-0773#
*Jun  4 00:17:19.830: %VLAN-5-CONFIG_I: Configured from console by console
SW1-AF-2025-0773#
  
```

The screenshot displays a Kali Linux VM environment. In the foreground, a terminal window titled "Kali Linux 1" shows the execution of several commands:

```

root@kali:~# cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
root:x:0:0:root:/root:/bin/sh

root@kali:~# sudo su -
root@kali:~# cd /opt
root@kali:~# ls -la
total 12
drwxr-xr-x 3 root root 4096 Jan 28 18:17 .
drwxr-xr-x 1 root root 4096 Jan 28 18:17 ..
drwxr-xr-x 2 root root 4096 Jan 28 18:17 solarwinds

root@kali:~# cd /opt/solarwinds
root@kali:~# ./solarwinds.sh
[+] Running Clean Attack - Adding Packet (2025-W773)
[*] [INTERFACE: eth0 | MAC: 08:00:27:1d:a3:b0 ] Privileged: 0
[-] BPOD enabled: 0x1

```

In the background, another terminal window shows the output of the SolarWinds Solar-Putty tool, displaying the configuration of the GigabitEthernet0/1 interface on SW1-AF-2025-W773. The configuration includes setting the interface to non-trunking mode, disabling BPDU guard, and showing the current status of the interface as err-disabled.

```

Portfast has been configured on GigabitEthernet0/1 but will only have effect when the interface is in a non-trunking mode.
SW1-AF-2025-W773(config-if)#exit
SW1-AF-2025-W773(config)#end
*Jun  4 00:34:55.836: %SPANTRIE-2-BLOCK_BPDUGUARD: Received BPDU on port Gi0/1 with BPDU Guard enabled. Disabling port.
SW1-AF-2025-W773#
*Jun: 4 00:34:55.838: MMH-4-ERR_DISABLE: bpduguard error detected on Gi0/1, putting Gi0/1 in err-disable state
*Jun: 4 00:34:56.908: ALIENPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
*Jun: 4 00:34:37.181: SVGS-S-COMPILG-I: Configured from console by console
*Jun: 4 00:34:37.803: ALINK-S-UPDOWN: Interface GigabitEthernet0/1, changed state to down
SW1-AF-2025-W773#show spanning-tree vlan 10

VLAN0010
Spanning tree enabled protocol ieee
Root ID    Priority    32768
Address    0c0d.8a96.0000
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID   Priority    32768 (priority 32768 sys-id-ext 10)
Address     0c0d.8a96.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 15 sec

Interface      Role Sts Cost      Prio Nbr Type
-----
Gi0/8          Desg FWD 4       128.1 P2p
Gi0/7          Desg FWD 4       128.3 P2p

SW1-AF-2025-W773#show interfaces GigabitEthernet0/1 status
Port      Name        Status      Vlan      Duplex  Speed Type
Gi0/1     err-disabled err-disabled 10         auto    auto R145
SW1-AF-2025-W773#

```

6. Contra-medidas

Contra-medida	Comandos aplicados	Resultado verificado
BPDU Guard (aplicado)	(Gi0/1)# spanning-tree portfast (Gi0/1)# spanning-tree bpduguard enable	Puerto Gi0/1 entro en err-disabled al recibir el primer BPDU de Kali
Root Guard	(Gi0/2)# spanning-tree guard root	Puerto bloqueado si recibe BPDU superior al root actual
Fijar Root Bridge	spanning-tree vlan 10 priority 4096	SW1 siempre gana la eleccion STP

6.1 Verificacion aplicada

```
SW1(config)# interface GigabitEthernet0/1
SW1(config-if)# spanning-tree portfast
SW1(config-if)# spanning-tree bpduguard enable
SW1(config-if)# exit

# Resultado inmediato tras enviar BPDUs:
SW1# show interfaces GigabitEthernet0/1 status
# Gi0/1 err-disabled 10

SW1# show spanning-tree vlan 10
# This bridge is the root (SW1 recupero el rol)
```